National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Electronic Devices & Circuits	Course Code:	EE225
Program:	Electrical Engineering	Semester:	Fall 201
Duration:	180 Minutes	Total Marks:	100
Paper Date:	18-12-2017	Weight	50%
Section:	ALL	Page(s):	4
Exam Type:	Final		

Student : Name:______________________ Roll No.______________ Section:____

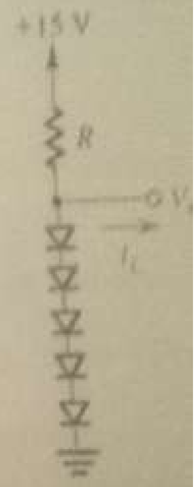
Instruction/Notes: Carefully read the paper and attempt each question (only once). Show all calculations, direct answers are not acceptable. Your answers should be approximated up to two decimal places. Attempt all parts of a question together. Take your assumption if any parameter is missing.

Question: 1 (CLO # 1) **10 Marks**

Design the circuit shown in figure so that $V_o = 4V$ when $I_L = 0$ and V_o changes by 40 mV per 1 mA of load current.
Find:

a) The value of R
b) The junction area of each diode relative to a diode with 0.7 V drops at 1mA current. Assume all diodes are identical and $n = 1$.

Hint: the junction area is directly proportional to saturation current I_s.

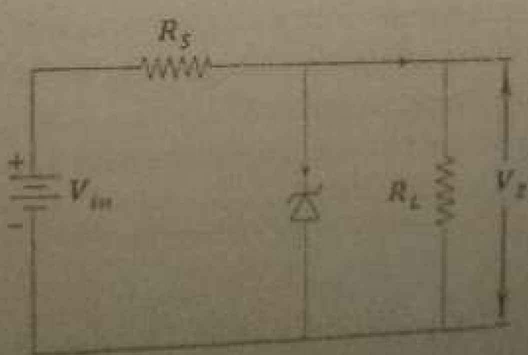


Question: 2 (CLO # 2) **15 Mar**

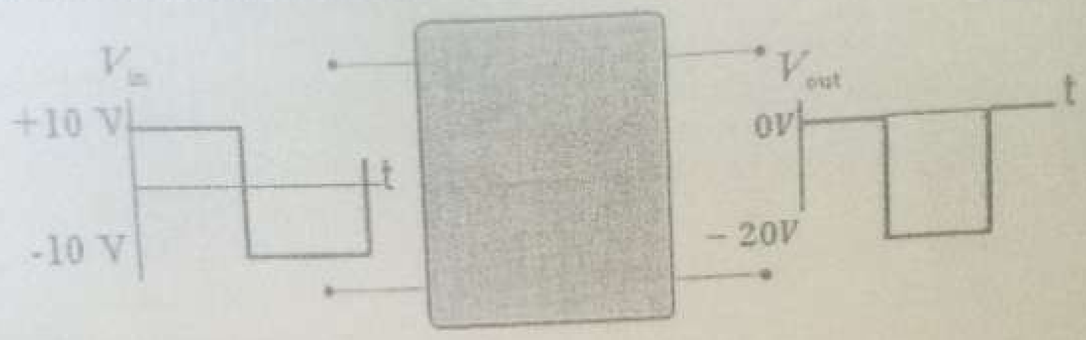
a) Consider the following circuit

I. Assume $V_{in} = 15V$, $R_s = 1k\Omega , V_z = 5V$ and $I_{knee} = 1mA$, Find the minimum valu of R_L so that diode stays in breakdown region.

II. Assume $V_{in} = 15V$, $R_L = 200\Omega , V_z = 5V$ and $I_{knee} = 1mA$, Find the maximum va of R_s so that diode stays in breakdown region.



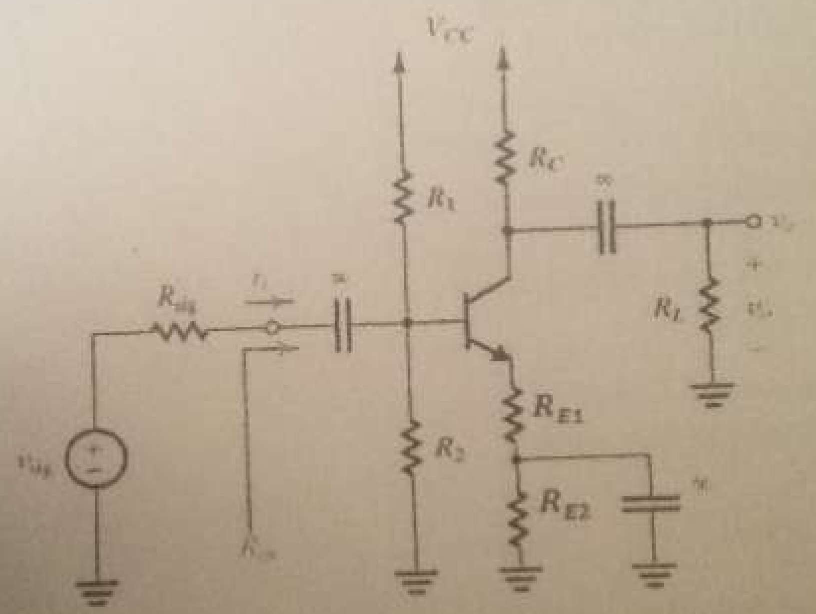
b) Draw the circuit inside the black box.



35 Marks

Question: 3 (CLO # 4,5)

Consider the following amplifier circuit with
$R_1 = 51k\Omega$, $R_2 = 32.9k\Omega$, $R_C = 4k\Omega$, $R_{E1} = 1k\Omega$, $R_{E2} = 3k\Omega$, $V_{CC} = 12V$
$R_{sig} = 50\Omega$, $v_{sig} = 1V_{peak}$ and $R_L = 1k\Omega$, $\beta = 100$



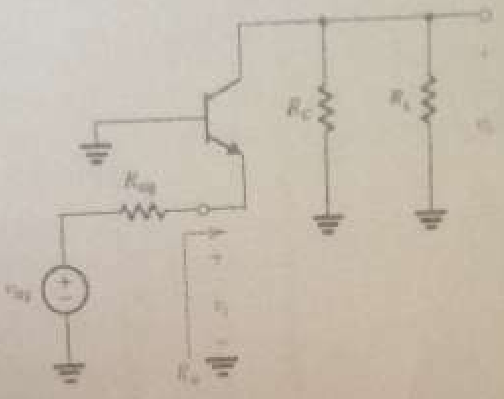
a) Find V_B, V_E, V_C, I_E and I_C.
b) Draw small signal equivalent model of the circuit.
c) Find small signal parameters?
d) Find the voltage gain $\frac{v_o}{v_{sig}}$, input resistance and output resistance of the amplifier?
e) If we replace both emitter resistors with a single resistor of $4k\Omega$ with capacitor connect to emitter directly, what is the effect on voltage gain, input and output resistances? Ver your results by calculating these parameters?
f) Comment which one is a better voltage amplifier (with or without split R_E) and why?

uestion: 4 (CLO # 5) 15 Marks

a) A common base amplifier is required to amplify a signal delivered by a co-axial cable having a source resistance of 50Ω.

I. What bias current I_C should be utilized to obtain R_{in} that is matched to the cable resistance?

II. To obtain an overall voltage gain of G_V of 40 V/V, what should the total resistance in the collector be?



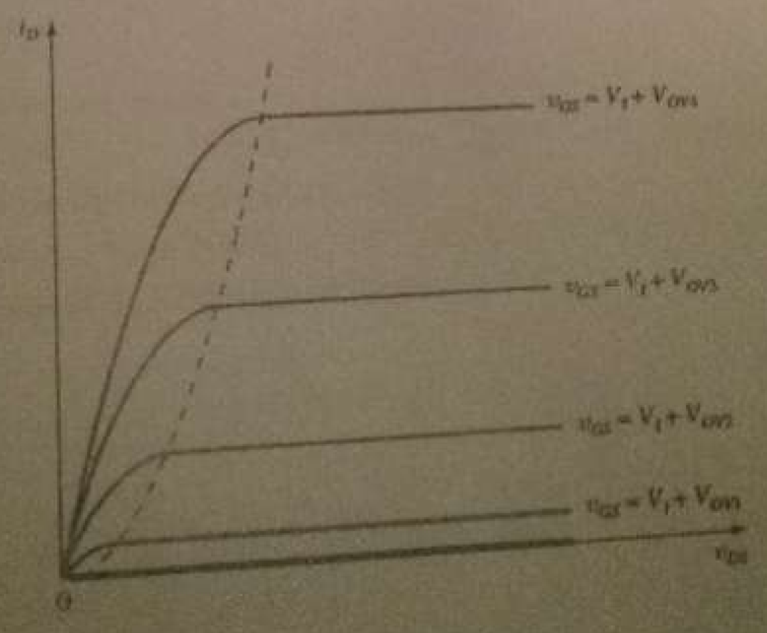
b) Compare common emitter, common base and common collector amplifiers? Use voltage gain, input resistance and output resistance as analysis parameters.

Question: 5 (CLO # 6) 15 Marks

a) The $i_D - v_{DS}$ characteristics of an n-channel enhancement MOSFET are given in the figure below.

i. Identify the three regions of operation of the NMOS on the characteristic curve.

ii. Give the Operating conditions for V_{DS} and V_{GS} and expression for current I_D in each region.



b) Draw and label the physical structure of a PMOS transistor.

uestion: 6 (CLO # 7) **10 Marks**

Design the given circuit with following requirements:

$V_{DD} = +5V$, $I_D = 0.32mA, V_S = 1.6V, V_D = 3.4V$ with a $1\mu A$ current through the voltage divider R_{G1}, R_{G2}. Let $V_t = 1V$ and $k'_n\left(\frac{W}{L}\right) = 1mA/V^2$. Neglect the channel length modulation effect.